Module 5

1. What are the key features of Wi-Fi 6 , 6E and 7 and how do they differ from previous standards like WiFi 5 (802.11ac) ?

Feature	Wi-Fi 5 (802.11ac)	Wi-Fi 6 / 6E (802.11ax)	Wi-Fi 7 (802.11be)
Frequency Band	5 GHz only	2.4 & 5 GHz (6 GHz in 6E)	2.4, 5 & 6 GHz
Modulation	256-QAM	1024-QAM	4096-QAM
Channel Width	Up to 160 MHz	Up to 160 MHz	Up to 320 MHz
Multi-user Tech	MU-MIMO (DL only)	OFDMA + MU-MIMO (UL & DL)	Enhanced OFDMA + MU-MIMO
New Features	—	BSS Coloring, TWT	MLO, rTWT, Preamble Puncturing
Performance	High speed	Better efficiency & low latency	Ultra-high speed & very low latency

2. Explain the role of OFDMA in Wi-Fi 6 and how it improves network efficiency.

Role of OFDMA:

- Divides a channel into smaller parts called Resource Units (RUs)
- The Access Point (AP) assigns these RUs to different devices
- Supports both uplink and downlink communication

How it improves efficiency:

- Multiple devices can transmit simultaneously , reduces waiting time
- Reduces collisions compared to CSMA/CA
- Better utilization of channel bandwidth
- Lower latency, especially in dense networks
- Efficient for small data transmissions (IoT devices)

3. Discuss the benefits of TWT in Wi-Fi 6 for IoT devices.

- saves battery life: devices sleep most of the time and wake only when needed
- Reduces unnecessary communication: no need to constantly check for data
- Lower power consumption: ideal for battery-operated devices like sensors and wearables
- Reduces network congestion: fewer devices active at the same time
- Improves efficiency: scheduled communication avoids collisions

4. Explain the benefits of the 6 GHz frequency band in Wi-Fi 6.

- **More available channels:** large spectrum so less congestion
- **Less interference:** fewer legacy devices use 6 GHz
- **Supports wider channels:** up to 160 MHz so higher speed
- **Better performance in dense areas:** suitable for offices, campuses
- **Lower latency:** faster and more stable communication

5. Compare and contrast Wi-Fi 6 and Wi-Fi 6E in terms of range, bandwidth, and interference.

Feature	Wi-Fi 6 (802.11ax)	Wi-Fi 6E
Frequency Band	2.4 GHz & 5 GHz	2.4, 5 & **6 GHz**
Range	Better (especially 2.4 GHz)	Shorter (6 GHz has less penetration)
Bandwidth	Up to 160 MHz	Up to 160 MHz
Interference	More (crowded 2.4 & 5 GHz)	Very low (6 GHz is less congested)
Performance	Good in mixed environments	Better in dense environments

6. What are the major innovaΘons introduced in Wi-Fi 7 ?

- 4096-QAM (higher data rate)
- 320 MHz channel bandwidth
- Multi-Link Operation (MLO) – uses multiple bands at same time
- Enhanced OFDMA and MU-MIMO
- Supports up to 16 spatial streams
- Lower latency with shorter guard intervals
- Improved scheduling and resource allocation
- Better QoS and traffic prioritization
- Smarter aggregation and retransmission

7. Explain the concept of MLO and its impact on throughput and latency.

Multi Link Operation is a feature in Wi-Fi 7 where a device can connect to multiple frequency bands (2.4 GHz, 5 GHz, 6 GHz) at the same time instead of using only one link.

- Device uses multiple links simultaneously
- Data can be split and sent across different bands
- AP manages and coordinates these links

Impact on throughput:

- Increases overall data rate (more bandwidth used)
- Combines multiple links so higher speed
- Improves reliability (backup links available)

Impact on latency:

- Reduces delay by sending data through the fastest link
- Avoids congestion by switching links
- Provides smoother performance for real-time applications

8. What is the purpose of 802.11 k? How do they aid in roaming?

802.11k is used for Radio Resource Management. It helps a Wi-Fi client (device) find the best nearby Access Point (AP) for roaming.

How it aids in roaming:

- AP provides a Neighbor Report (list of nearby APs)
- Includes details like:
    - Channel
    - BSSID
    - Signal strength
- Client uses this information to choose the best AP
- No need to scan all channels → faster decision
- Faster roaming

- Reduced scanning time
- Better performance in dense networks

9. Explain the concept of Fast BSS TransiƟon (802.11r) and its benefits in mobile environments.

802.11r enables fast roaming between Access Points (APs) by reducing authentication delay. It allows the client to pre-authenticate with the next AP and reuse security keys (Fast Transition keys) before moving.

How it works:

- Client prepares connection with target AP in advance
- Uses Fast Transition (FT) key for quick authentication
- Handoff happens quickly without full re-authentication

Benefits in mobile environments:

- Very fast handoff between APs
- Reduces delay and packet loss
- Prevents call drops in VoIP/video calls
- Smooth user experience while moving
- Ideal for real-time applications (voice, video, AR/VR)

10. How do 802.11 k/v/r work together to provide seamless roaming in enterprise networks?

- **802.11k (Discovery):**
  Provides neighbor report , client knows nearby APs
  Helps choose best AP quickly
- **802.11v (Decision):**
  AP guides/steers client to a better AP
  Improves load balancing and roaming decision
- **802.11r (Execution):**
  Performs fast handoff using pre-authentication
  Reduces delay during switching